

On-Chain Parametric Cost Insurance for Underserved Latin-American Digital Workers

Record of Closed Iterations and Request for Methodological Guidance

d² finance

github.com/wvs-finance

2026-05-06

§1. Motivation, problem, suggested solution, challenges

Motivation. We study Latin-American digital workers paid in Colombian pesos who carry recurring USD-denominated professional-input obligations (subscription AI services, API usage fees, cloud compute, software-as-a-service spend), a population whose macro-risk exposure is dominated by FX volatility: for the Colombian case specifically, the FX market is the efficient aggregator of macro shocks, with TES yields and core inflation operating as followers of the COP/USD path (Rincón-Torres et al., 2021, 2023; Rincón-Castro et al., 2021; Fuentes et al., 2014); the same FX-as-primary-transmission pattern is documented across emerging markets generally (Calvo and Reinhart, 2002; Rey, 2015; Bruno and Shin, 2015).

Problem. When the COP weakens against the USD, the USD-denominated cost line translates into unbudgeted COP expenditure. Existing risk-management practice for this population is *indirect*: USD hoarding via neo-bank dollar-savings products (e.g., Littio, Global66, Nubank’s USD wallet pilot) and informal hard-currency holdings, rather than a direct convex hedge on the COP/USD path. To the project’s knowledge, no Colombian retail-bank or fintech product currently offers a parametric FX-call-option-style cost-insurance contract at the hundred-to-low-thousands-USD monthly notional these workers carry; the indirect substitutes charge no explicit premium for convexity and provide no claim-on-realisation against an exchange-rate threshold. The supervisor is invited to flag any product the project has missed in this scan.

Suggested solution. We identify the macro risks X that quantitatively produce these cost shocks, validate them empirically, and design parametric cost-insurance products that compensate for the FX-pass-through exposure (Carter et al., 2017; Mahul and Stutley, 2010; Clarke, 2016). The demand pattern we target — underserved-country households purchasing depreciation insurance in the form of a call option on the foreign-exchange rate — is documented in a recent practitioner interview (Leifke, 2026): “the instrument that people really care about, especially in emerging markets, is more of like a devaluation insurance

policy ... call options are kind of perfect for this because you don't need a credit relationship. You just pay a premium upfront." The intermediation product compensates the worker in a unit she actually spends (COP, or a COP-pegged purchasing-parity stablecoin); the implementation venue is on-chain settlement infrastructure (described in the 37-page technical companion §4), not the economic primitive.

Challenges.

- Identification posture — predictive versus causal (see §4.2).
- Convex-payoff sufficiency under insurance-underwriting standard (see §4.2).
- Multi- Y differential calibration; familywise versus false-discovery correction across legs (see §4.2).
- Anti-fishing protocol defensibility — pre-registration thresholds $N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$ (see §4.2).
- Cross-iteration familywise error and iteration-selection bias (see §4.2).

Bridge to §2. We have run four iterations to date, all closed or parked; the model specifications, results, and failure post-mortems follow in Section 2. (A fifth iteration — Pair D, on Colombian young-worker employment in the broad services aggregate — closed PASS but is omitted here on population-orthogonality grounds: its target population is BPO young-worker employment, not the digital-worker FX-pass-through cost-shock population in scope for the present letter; that iteration's record is preserved in the 37-page technical companion and in the project memory.)

§2. Four iterations

dev-AI Stage-1 — Section J ICT through COP/USD lag (FAIL). Colombian young workers (14–28) restricted to CIU Rev. 4 Section J (Information and Communication, Divisions 58–63), with monthly cadence, **FEX_C_2018** expansion factors, and the lag panel detailed below.

$$Y_t = \alpha + \beta_6 X_{t-6} + \beta_9 X_{t-9} + \beta_{12} X_{t-12} + \varepsilon_t$$

where $Y_t = \text{logit}(\text{share}_t^J)$ and $X_{t-k} = \log(\text{COP/USD}_{t-k})$; pre-registered $H_1 : \beta_{\text{composite}} > 0$. **Data.** Source: DANE GEIH micro-data restricted to CIU Rev. 4 Section J (Divisions 58–63); Banrep TRM. **Frequency:** monthly. **Window:** 2015-01-31 through 2026-03-31. **Sample:** $N = 134$ monthly. **Transformations:** logit on Section-J share; log-level on FX; same lag panel as Pair D; HAC($L = 12$); **FEX_C_2018**; empalme residual-bias diagnostic at the 2020-12 → 2021-01 Marco-2018 boundary. **Repo:** notebooks/dev_ai_cost. **Spec sha:** 7c72292516...751f5a. **Result.** $\hat{\beta}_{\text{composite}} = -0.14613$, HAC SE 0.0847, $t = -1.726$, one-sided $p \approx 0.958$ against $H_1 : \beta > 0$; R-classifier MIXED (3 of 4 R-rows agree on negative sign). **Interpretation.** FAIL — sign-flipped at Section-J narrow cut. **Why it fails.** The Section-J substrate does not exhibit

the lagged-FX-pass-through-to-employment pattern the BPO-offshoring narrative pre-supposed; an empalme-residual bias surfaced (the Phase-1 boundary diagnostic returned a $+0.375$ logit- Y jump at the 2020-12 $\rightarrow$ 2021-01 Marco-2018 boundary against a ± 0.335 envelope), so the post-2021 level shift is not fully neutralized at the Section-J cut. The R2 Section-M sensitivity arm (CIU Sections 69–75) returned $\hat{\beta}_{\text{composite}} = +0.45482801$, $t = +4.73$, $p = 1.13 \times 10^{-6}$, but this is a not-pre-registered candidate-next-iteration object, not a graduated PASS, and the rejection at Section-J falsifies the per-user instrument design that the dev-AI-paying population would have hosted.

FX-vol-CPI surprise — Colombian CPI through TRM realised volatility (FAIL). Colombian corporates and remittance recipients with FX-volatility exposure on COP/USD around DANE CPI publication, weekly cadence on a Friday anchor, with Y_w the weekly realised volatility of the Banrep TRM daily log-return panel and the regressor the lagged signed AR(1) CPI surprise.

$$RV_w^{\text{COP/USD}} = \alpha + \beta \text{Surp}_{w-1}^{\text{CPI}} + \varepsilon_w$$

where RV_w is realised volatility in week w and $\text{Surp}_{w-1}^{\text{CPI}}$ is the lagged signed monthly CPI surprise mapped to weekly cadence (carried across the release week, zero otherwise), with a small set of macro-surprise controls in the primary. **Data.** **Source:** Banrep TRM daily log-return panel; DANE CPI release calendar + realized values for AR(1) surprise construction. **Frequency:** weekly (Friday anchor). **Window:** 2008-01-02 through 2026-02-23. **Sample:** $N = 947$ weekly. **Transformations:** weekly RV from daily log-returns; signed AR(1) CPI surprise mapped to weekly cadence (carried across release week, zero otherwise); HAC(4) Newey-West SE; small set of macro-surprise controls. **Repo:** [notebooks/fx_vol_cpi_surprise](#). **Spec fingerprint:** `nb1_panel_fingerprint.json`. **Result.** $\hat{\beta}_{\text{CPI}} = -0.000685$, HAC(4) SE 0.001794, 90% interval $[-0.003636, +0.002266]$ contains zero; gate criterion $\hat{\beta} - 1.28 \cdot \widehat{\text{SE}} = -0.002981 < 0$ fires the FAIL. **Interpretation.** FAIL — no weekly-RV response to CPI surprises. *Why it fails and sensitivity caveat.* The pre-registered T1 exogeneity test rejects at $F = 15.12$, $p \approx 1.3 \times 10^{-9}$, so the iteration is a predictive regression rather than a structural-causal one and $\hat{\beta}$ cannot be read as an impulse-response. Two pre-registered sensitivities returned positive significant 90% intervals at the one-sided $\alpha = 0.05$ convention — A1 monthly cadence $\hat{\beta} = +0.0152$ on $[+0.0057, +0.0246]$; A4 release-day-excluded $\hat{\beta} = +0.0033$ on $[+0.0005, +0.0062]$ — but neither is the iteration's main finding: the pre-registered weekly primary is the result, and A1/A4 sit as candidate inputs to a possible future monthly-cadence-primary specification rather than as a rescue of the weekly primary.

Phase-A.0 Remittance — Colombian remittance corridor through FX (EXIT_NON_REMITTAN Colombian household remittance recipients indexed against the Colombian remittance corridor with FX pass-through, with Y_w inherited from the FX-vol weekly panel and X_{w-1} a candidate on-chain remittance-flow proxy constructed from cCOP and COPm aggregate user activity on Celo.

$$RV_w^{\text{COP/USD}} = \alpha + \beta \Delta \text{Remit}_{w-1}^{\text{Col}} + \varepsilon_w$$

where $\Delta\text{Remit}_{w-1}^{\text{Col}}$ is the lagged Colombian household-remittance flow change as proxied on Celo on-chain activity, partitioned to attempt to isolate household remittance from third-party-DEX activity, treasury roundtripping, and bot arbitrage. *Data.* **Source (planned):** Ban-rep monthly remittance + Celo on-chain cCOP / COPm aggregate user activity. **Frequency:** weekly (planned; never reached final N). **Window:** 2024-09 through 2026-04 (planned). **Sample:** not produced — exited pre-estimation. **Transformations:** pre-registered partition rule for household remittance vs third-party-DEX activity, treasury roundtripping, bot arbitrage; never executed. **Spec:** [docs/specs/2026-04-20-remittance-surprise-trm-rv-design.md](#) (no notebook directory — spec exited pre-estimation). **Spec chain:** Phase-A.0 Rev 4.1 (kill-criterion-bearing). *Result.* No empirical estimate produced — exited on kill criteria pre-estimation 2026-04-24. *Interpretation.* EXIT — never estimated; killed by k1 + partial-k2. *Why it exits.* Task 11.F Axis-1 event research returned zero of thirty peak-day flows fingerprinting as remittance, with approximately 87% of surveyed flow decomposing into TRM arbitrage, treasury roundtripping, campaigns, and bot activity; the cCOP/COPM aggregate user activity on Celo is third-party-DEX volume, not Colombian household remittance, so no post-hoc filter could reconstruct an X-driver that was never measured upstream. The exit is load-bearing on the kill-criterion discipline: absent the pre-registered EXIT, the rev-4 plan's non-stop filter iteration would have run to a filter that appeared to “rescue” a signal that was never captured.

P1 Bittensor SN18 — event-study apparatus (PARKED). The proprietary-AI policy-shock-exposed population (LATAM developers paying USD-denominated AI APIs as the downstream candidate; the apparatus pre-dates target-population narrowing), with Y_t a Bittensor SN18 (Cortex.t) alpha event-study indicator and X_t a set of pre-pinned policy-event dates (Bittensor halving 2025-12-15; dTAO mainnet activation 2025-02-13; Cortex.t milestones).

$$\text{AR}_{i,\tau} = \alpha_i + \sum_{\tau=-L}^{+L} \gamma_{\tau} \mathbf{1}\{t_i + \tau \in \mathcal{E}\} + \varepsilon_{i,\tau}$$

where $\text{AR}_{i,\tau}$ is the event-window abnormal alpha indicator at relative time τ for event i , $\mathcal{E}$ is the pre-pinned policy-event set, and the apparatus is evaluated over a nine-cell verdict cube (verdict-eligibility crossed with robustness-consistency), Bonferroni-corrected at $\alpha_{\text{primary}} = 0.0167$ over the conjunctive family P1+P2+P3 with $N_{\min}^{\text{events}} = 8$ and an asymmetric placebo gate. *Data.* **Source:** Bittensor SN18 (Cortex.t) alpha event-study; pre-pinned policy-event dates (Bittensor halving 2025-12-15; dTAO mainnet 2025-02-13; Cortex.t milestones). **Frequency:** event-window (asymmetric L). **Window:** per-event $\pm L$. **Sample:** planned $N_{\min}^{\text{events}} = 8$; data ingest deferred. **Transformations:** event-window abnormal-alpha indicator; nine-cell verdict cube; Bonferroni at $\alpha_{\text{primary}} = 0.0167$ over P1+P2+P3. **Spec:** [docs/specs/2026-04-27-p1-sn18-event-study-design.md](#) (no notebook directory — apparatus parked pre-ingest). **Spec sha:** f855e036d3...b47aeab. *Result.* None — apparatus parked for the record on 2026-04-27; no graduating empirical claim produced. *Interpretation.* PARKED — apparatus complete, no claim graduated. *Why parked.* Apparatus complete; no graduating empirical claim. The user pivoted to the simple-composite- β fast-lane track (Pair D, dev-AI Stage-1) and parked P1 to preserve the intellectual capital of the spec work, with re-activation conditions documented in the parking memory.

§3. Risk-target framing

The risk target is FX-volatility exposure for a specific population: Latin-American digital workers paid in Colombian pesos (COP cash-flows) but carrying recurring USD-denominated professional-input obligations. When the COP weakens against the USD, the USD-denominated cost line translates into an unbudgeted COP-expenditure increase. The intermediation product target is a contract that compensates for this FX-pass-through cost shock, denominated in a unit the worker actually spends (COP, or a COP-pegged purchasing-parity stablecoin). The Abrigo Lemma’s abstract $Y_{\text{inequality}}(t) = R_a(t) - R_c(t)$ framing — present in the 37-page technical companion §3 — is the eventual analytical generalisation; the entry-point letter introduces the concrete target only.

§4. Asks

§4.1 Suggested specifications

We invite the supervisor to suggest the (Y, X) pair, lag structure, sample window, primary inference convention, and robustness arms for the next iteration. Specifically: which sectoral cut should follow the dev-AI Section J FAIL given the R2 Section M sensitivity signal (CIU Rev. 4 Sections 69–75: consultancy, scientific, administrative); whether to repeat the FX-vol-on-CPI-surprise primary at A1 monthly cadence as the next pre-registration; or whether to pivot to a non-Colombian LATAM panel.

§4.2 Risks

The bullets below are risks the project has already identified; we ask the supervisor to add risks we have not flagged.

- Identification posture — predictive-regression versus structural-causal; T1 exogeneity rejected on FX-vol; HAC fixed- L per iteration.
- Convex-payoff sufficiency under insurance-underwriting standard — mean- $\hat{\beta}$ is necessary but not sufficient; [Carter et al. \(2017\)](#), [Mahul and Stutley \(2010\)](#), and [Clarke \(2016\)](#) set the standards.
- Multi- Y differential calibration — joint-leg correlation, FWER versus FDR, regime-switching.
- Anti-fishing protocol defensibility — $N_{\text{MIN}} = 75$, $\text{POWER}_{\text{MIN}} = 0.80$, $\text{MDES}_{\text{SD}} = 0.40$ inherited from Phase-A.0 Rev-5.3.1, not re-derived.
- Cross-iteration FWER and iteration-selection bias — five closed iterations across the project roster (the four presented here plus Pair D, omitted on population-orthogonality grounds); meta-multiple-comparison concern; [Heckman \(1979\)](#) as candidate frame.
- Stambaugh small-sample bias on persistent-regressor predictive regressions — [Stambaugh \(1999\)](#), [Phillips \(2014\)](#); log-FX is near-unit-root at monthly frequency.

- Empalme residual bias — DANE Marco-2018 methodology break not fully neutralised; surfaced in dev-AI Stage-1 NB02 Trio 1 boundary anomaly.

§4.3 Bibliography

The current bibliography is documented in the 37-page companion’s Appendix B and is reproduced as `refs.bib`. We invite the supervisor to recommend additional references he considers load-bearing for the framework’s empirical work, particularly on (a) parametric-index insurance pricing in low-counterparty-credit-standing populations, (b) sectoral employment-share predictive regressions on real-effective-exchange-rate panels, and (c) cross-iteration false-discovery-rate correction regimes for observational pre-registration.

§4.4 Connections

We invite the supervisor to introduce the project to other researchers, practitioners, or institutions whose work might complement this iteration roster — including but not limited to parametric-insurance underwriters with LATAM exposure, FX-derivatives operators serving sub-corporate notionals, on-chain settlement engineers, and applied econometricians working on premature deindustrialization in Latin America. The framework’s empirical work is open to collaboration; the supervisor’s network is the most efficient channel for identifying complementary contributors.

References

- Valentina Bruno and Hyun Song Shin. Cross-border banking and global liquidity. *The Review of Economic Studies*, 82(2):535–564, 2015.
- Guillermo A. Calvo and Carmen M. Reinhart. Fear of floating. *The Quarterly Journal of Economics*, 117(2):379–408, 2002.
- Michael R. Carter, Alain de Janvry, Elisabeth Sadoulet, and Alexandros Sarris. Index insurance for developing country agriculture: A reassessment. *World Development*, 94:199–216, 2017.
- Daniel J. Clarke. A theory of rational demand for index insurance. *American Economic Journal: Microeconomics*, 8(1):283–306, 2016.
- Miguel Fuentes, Pablo Pincheira, Juan Manuel Julio, Hernán Rincón, Santiago García-Verdú, Miguel Zerecero, Marco Vega, Erick Lahura, and Ramon Moreno. The effects of intraday foreign exchange market operations in Latin America: results for Chile, Colombia, Mexico and Peru. Borradores de Economía 849, Banco de la República, Bogotá, 2014. Co-indexed as BIS Working Paper No. 462.
- James J. Heckman. Sample selection bias as a specification error. *Econometrica*, 47(1):153–161, 1979. doi: 10.2307/1912352.
- Robert Leifke. Builder stories: Inside numo. Interview, Atrium Academy (YouTube), <https://www.youtube.com/watch?v=-nPTjKRMSK8&t=1115s>, March 2026. Streamed live 2026-03-18.

Quote on emerging-market demand for FX-depreciation insurance as a call option at 18:35–19:20.

Olivier Mahul and Charles J. Stutley. *Government Support to Agricultural Insurance: Challenges and Options for Developing Countries*. World Bank, Washington, DC, 2010.

Peter C. B. Phillips. On confidence intervals for autoregressive roots and predictive regression. *Econometric Theory*, 30(3), 2014.

Hélène Rey. Dilemma not trilemma: The global financial cycle and monetary policy independence. NBER Working Paper No. 21162, 2015.

Hernán Rincón-Castro, Norberto Rubiano-López, Julián Yaya-Garzón, and Héctor Zárate-Solano. Traspaso de la tasa de cambio a la inflación básica en Colombia. Borradores de economía, Banco de la República, Bogotá, 2021.

Andrés Rincón-Torres, Nicolás Rojas-Silva, and Juan Manuel Julio-Román. The interdependence of FX and Treasury bonds markets: The case of Colombia. Borradores de Economía 1171, Banco de la República, Bogotá, 2021.

Andrés Rincón-Torres, Daniel De la Hortúa-Pulido, Nicolás Rojas-Silva, and Juan Manuel Julio-Román. The low frequency effect of macroeconomic news on Colombian government bond yields. Borradores de economía, Banco de la República, Bogotá, 2023.

Robert F. Stambaugh. Predictive regressions. *Journal of Financial Economics*, 54(3), 1999.